Extracting Social and Clinical Impacts of Substance Use from Reddit with Structured Pipelines and Ensemble Search

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Abstract

We present our submission to SMM4H–HeaRD 2026 Task 7, which extracts *Clinical* and *Social* impacts of substance use from Reddit posts as token-level BIO tagging. The benchmark is hard for two reasons: the label prior flips between train (≈75% Clinical) and test (≈70% Social), and many entities are implicit colloquialisms without medical vocabulary. We compare five model families—BiLSTM–CRF, GLiNER, SocBERT, StressRoBERTa, and DeBERTa-v3-large—under a consistent LR sweep, organise our contributions into four buckets (imbalanced-inputs training, advanced regularisation, architecture, and ensemble techniques), and treat preprocessing as an orthogonal raw-vs-cleaned experimental axis. Our best single model, definition prompting, reaches relaxed F1 $= 0.613$; our best system, a four-model majority-vote ensemble, reaches 0.635, beating the published baseline on the same dev split by $+0.025$. On the official test leaderboard our submitted system (an earlier configuration than the dev-best ensemble) placed **5th of 11** teams (strict F1 $= 0.49$, relaxed F1 $= 0.45$). The main lesson: training-objective diversity combined with hard majority voting is a stronger lever than polishing any single model.

1 Introduction

Public-health surveillance on social media has become an important source of population-level signal for substance-use disorders (Sarker et al., 2019). The SMM4H–HeaRD 2026 shared task Track 7 asks systems to extract, from opioid-subreddit posts, two kinds of consequences that users describe in natural language: *ClinicalImpacts* (physiological and psychological effects such as overdose or withdrawal) and *SocialImpacts* (occupational, legal, and relational consequences such as losing a job or getting arrested). The task is formulated as token-level NER with five BIO tags: `O`, `B/I-ClinicalImpacts`, `B/I-SocialImpacts`.

What makes this benchmark unusually difficult is a striking mismatch between the training and test distributions. The training split is dominated by Clinical entities (≈75%), and the test split flips to Social (≈70%). On top of that, many impacts are expressed *implicitly*—"those weeks were hell" is a Clinical span without a single clinical token—so lexical and ontology-based cues are weak. The published baseline on Reddit-Impacts 2.0 reaches relaxed F1 $= 0.610$, and inter-annotator agreement is around 0.81 (Dey et al., 2025), leaving a 0.20 gap that no single well-known recipe closes.

We treat this as a systems problem and report a thorough comparison across five model families under a uniform LR sweep and a paired *raw vs preprocessed* protocol. Our contributions are:

1. A disciplined baseline study across BiLSTM–CRF, GLiNER fine-tuned, SocBERT, StressRoBERTa, and DeBERTa-v3-large, showing that a general-purpose encoder dominates Reddit- and mental-health-adapted encoders on this task (§5.1).
2. A paired *raw vs preprocessed* protocol in which every model family is run in both a raw-text regime and a mojibake/URL/emoji-cleaned regime, so preprocessing is a first-class experimental variable rather than a confound (§4).
3. Four bucket-level contributions organised as: (i) imbalanced-inputs training (focal loss, definition prompting, multi-task auxiliary head, Recall-Boost); (ii) advanced regularisation (R-Drop, FGM + SWA); (iii) architecture (four structured-prediction pipelines — Hierarchical DeBERTa, Two-Step Impact, Sentence + Token, Span-nested GLiNER — plus domain-adapted backbones SocBERT and StressRoBERTa); and (iv) ensemble

techniques (weight-space model soup and prediction-space ensemble search) (§5).

4. An exhaustive ensemble search across both preprocessing regimes. A four-model majority-vote ensemble reaches relaxed F1 = **0.635** on the dev set, exceeding the published baseline by $+0.025$ on the same split (§5.2.4).

The clearest single takeaway from the study is that *training-objective diversity* is worth more than squeezing an additional point out of any individual recipe. Five of the six DeBERTa-based loss/regularisation recipes sit at or below the plain DeBERTa baseline when measured alone; they earn their keep by producing decorrelated errors that hard voting combines.

2 Related Work

Health-focused NER on social media. Extracting clinically-relevant spans from social-media text has been studied for medication mentions, adverse drug events, and more recently substance-use impacts (Sarker et al., 2019; Weissenbacher et al., 2022; Dey et al., 2025). The dominant approach is a transformer encoder with a token-classification head, sometimes augmented with domain adaptation. Reddit-Impacts 2.0 (Dey et al., 2025) is unusual in separating *clinical* and *social* outcomes and in deliberately flipping the label prior between train and test.

Domain-adapted encoders. Reddit-specific pre-training (Guo and Sarker, 2023) and mental-health-domain pre-training (Amalq, 2022) typically help when the downstream task shares vocabulary and register with the pre-training corpus. In our setting, however, the task labels are not register-bound and generic DeBERTa-v3 (He et al., 2023) outperforms both domain-adapted encoders.

Span-based NER and regularisation. GLiNER (Zaratiana et al., 2024) frames NER as open-type span scoring and is attractive for low-resource settings; we include both zero-shot and fine-tuned variants. Our advanced-regularisation bucket is built from off-the-shelf ingredients—focal loss (Lin et al., 2017), R-Drop (Liang et al., 2021), FGM adversarial training (Miyato et al., 2017), and SWA (Izmailov et al., 2018)—combined with model soup (Wortsman et al., 2022) and prediction-space ensembles (Kuncheva and Whitaker, 2003). Our contribution is the exhaustive cross-regime subset search over these ingredients.

Prior SMM4H iterations of the Reddit-Impacts track. The task first ran as SMM4H 2024 Task 4 on an earlier corpus version: UKYNLP (Obeidat et al., 2024) topped the leaderboard with encoder-LM ensembles, and Dolomites (Tortoreto and Mousavi, 2024) reached strict F1 = 0.208 with a Mistral-7B QLoRA setup plus in-domain augmentation and "drills" — evidence that even LLM-scale fine-tuning on augmented data leaves substantial strict-F1 headroom. At SMM4H–HeaRD 2025 (Klein et al., 2025) the top-ranked LLM Pros submission (Pradhan et al., 2025) used a prompt-engineered, fine-tuning-free LLM pipeline. Our work returns to the encoder-fine-tuning side of the design space and shows that an objective-diverse encoder ensemble can match those systems on the 2026 benchmark.

3 Task and Dataset

Reddit-Impacts 2.0. The dataset contains 1,378 posts from opioid-related subreddits, pre-split into train, dev, and test; the dev split has 258 documents, which means relaxed-F1 confidence intervals are wide (§9). The training and dev splits are both Clinical-heavy, while the test split flips to roughly 70% Social. Figure 1 visualises the split composition, the span-length distribution, and the fraction of posts that contain no entities at all (a non-trivial class, particularly on dev).

Evaluation. We report both Relaxed and Strict F1 over entity mentions. Relaxed F1 credits a prediction whose type is correct and whose span overlaps the gold span by at least one token; Strict F1 additionally requires exact span boundaries. Dev-set performance is the primary selection criterion because test-set labels are not available; we discuss the implications of this in §9.

4 Preprocessing as an Axis

A common pitfall in social-media NER is to conflate preprocessing with modelling: a single "cleaned" pipeline is used for every experiment and its effect is invisible. We treat preprocessing as a first-class experimental axis. Every family is run twice:

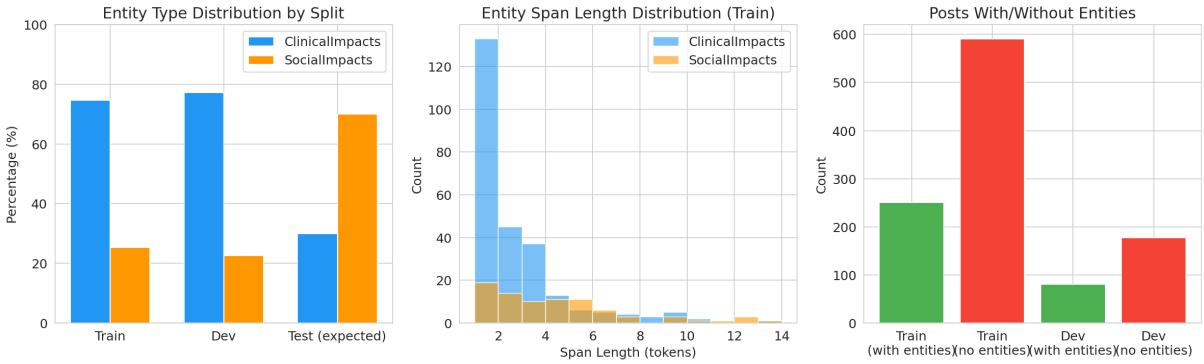

Figure 1: Reddit-Impacts 2.0 exploratory statistics. **Left:** Train/Dev are Clinical-heavy; Test flips to ≈70% Social. **Middle:** Social spans are longer and heavier-tailed than Clinical. **Right:** the majority of dev posts contain no entities at all, which makes a sentence-level presence head a useful auxiliary signal.

- **Raw regime.** Original Reddit text goes straight to the tokeniser. Usernames, URLs, mojibake (broken Unicode from earlier encoding/decoding trips), emoji, and stray punctuation are preserved. This regime maximises lexical fidelity but produces noisy token boundaries.

- **Preprocessed regime.** Four token-aligned operations: (i) mojibake normalisation (e.g. â → ’); (ii) in-place substitution of `@user`/`/u/user` mentions with `[USER]` and URLs with `[URL]`; (iii) deletion of emoji, punctuation-only tokens, and dataset metadata markers (`submission_title`, `submission_subreddit`); and (iv) post-hoc realignment at inference, where deleted positions are restored as `O` so the output BIO length matches the original tokenisation.

The effect of the regime is family-specific. BiLSTM–CRF and GLiNER both gain markedly from preprocessing (relaxed F1 $+0.045$ and $+0.061$ respectively). The plain DeBERTa baseline is actually *stronger* on raw text (0.595 vs 0.545), which is consistent with the hypothesis that large pre-trained encoders are already robust to Reddit-style noise and that aggressive cleaning removes genuine signal. Conversely, definition prompting and R-Drop only become their best selves in the preprocessed regime, because both rely on stable attention over a clean context. This asymmetry is exactly why treating preprocessing as an axis matters.

5 Methods

5.1 Model families

BiLSTM–CRF (sanity check). A frozen GloVe-300d embedding layer feeds a two-layer bidirectional LSTM (2×256 hidden), a linear projection to five tags, and a CRF with Viterbi decoding at inference time. Trained with AdamW (base LR $1\mathrm{e}{-3}$) for 30 epochs at batch size 30 and dropout 0.2.

GLiNER. We include two variants of GLiNER (Zaratiana et al., 2024): (a) fine-tuned on Reddit-Impacts with a 0.4 span-score threshold; and (b) a span-nested variant that enumerates nested candidate spans and applies overlap-aware decoding.

Domain-adapted encoders. SocBERT (Guo and Sarker, 2023) is pre-trained on Reddit posts; StressRoBERTa (Amalq, 2022) is pre-trained on mental-health text. Both are token classifiers with a linear head.

DeBERTa-v3-large (main backbone). He et al. (2023)’s disentangled-attention encoder with a linear token classification head is our primary backbone. All advanced innovations build on top of it.

5.2 Contribution buckets

Our strategy is organised as four bucket-level contributions. Most buckets build on top of DeBERTa; the domain-adapted-backbone half of bucket 3 is model-agnostic. The preprocessing axis (§4) runs orthogonally to every bucket and is not itself a bucket. Each DeBERTa-family recipe is swept under $\mathrm{lr} \in \{5\mathrm{e}{-4}, 2\mathrm{e}{-5}, 5\mathrm{e}{-5}\}$; GLiNER uses a safer sweep $\mathrm{lr} \in \{5\mathrm{e}{-6}, 1\mathrm{e}{-5}, 2\mathrm{e}{-5}\}$ because larger values destabilise span-score training. We

run the sweep in both regimes and report the best LR per family.

5.2.1 Bucket 1: Imbalanced-inputs training

Recipes in this bucket respond to class imbalance and the train→test prior flip, either by reshaping the loss, injecting label semantics, or sharing gradient with an auxiliary head that encodes the impact/no-impact boundary.

Focal loss. Standard focal loss (Lin et al., 2017) $\mathcal{L}_{\text{focal}}(p_t) = -\alpha_t (1-p_t)^{\gamma} \log p_t$ with $\gamma = 2.0$. The per-class weights α are data-driven: inverse training-frequency $\alpha_c \propto 1/f_c$, smoothed with exponent 0.3, renormalised to unit mean, and B/I-Social then scaled by 1.5 to push the model toward the under-represented Social class.

Definition prompting. We prepend a natural-language description of both entity types, separated by `[SEP]`, and mask the prefix from the loss with -100 labels so definitions are available to attention but contribute no gradient. Two prompt choices matter: we make the perspective explicit ("personally experienced by the post author"), biasing the model against third-person mentions, and list concrete exemplars per class drawn from the annotation guidelines (overdose, withdrawal, addiction, depression, . . . ; job loss, arrest, relationship breakdown, homelessness, . . .). This is the single best-performing recipe in the project.

Multi-task auxiliary head. We share the DeBERTa encoder between the token NER head and a sentence-level binary head at `[CLS]` that predicts *entity presence* $y = \mathbf{1}[\text{sentence has any B/I tag}]$. The combined loss is $\mathcal{L} = \mathcal{L}_{\text{token}} + 0.3 \cdot \mathcal{L}_{\text{presence}}$.

Recall-Boost (O-class downweighting). A pilot inspection of the plain baseline's dev-set predictions showed recall trailing precision, with 29 missed spans of which 13 were single-token implicit expressions. We respond by down-weighting the `O` class from 1.0 to 0.2 and up-weighting Social to 1.5, while keeping Clinical at 1.0. We combine this with the multi-task auxiliary head described above. §7 presents a fuller error analysis on the best single model.

5.2.2 Bucket 2: Advanced regularisation

R-Drop. Following Liang et al. (2021), we run each batch through the encoder twice with independent dropout masks and penalise disagreement with a symmetric KL. We also stack the multi-task sentence presence head, so the objective is $\mathcal{L} = \frac{1}{2}(\text{CE}(p_1) + \text{CE}(p_2)) + \alpha\, \text{KL}_{\text{sym}}(p_1 \| p_2) + 0.3\, \mathcal{L}_{\text{presence}}$, with $\alpha = 1.0$ and KL applied only on non-masked tokens. No new parameters are introduced.

FGM + SWA. Per step we perturb the `word_embeddings` by $w \leftarrow w + \varepsilon \cdot g/\|g\|$ with $\varepsilon = 0.5$ (Miyato et al., 2017), take a second forward pass, and sum clean and adversarial gradients before the optimiser step. After epoch 10 we compute a stochastic weight average (Izmailov et al., 2018) over remaining checkpoints as the final model.

5.2.3 Bucket 3: Architecture

Structural changes to the model: re-shaping the task into stages or swapping the encoder.

Structured prediction pipelines. Four structured variants:

Hierarchical DeBERTa. A `[CLS]` sentence classifier gates a multi-task NER: only sentences whose sentence head predicts "has-entity" are routed to the token head. During training we keep every positive sentence and only 10% of the no-impact sentences, so the token head is trained on a more balanced distribution.

Two-Step Impact Pipeline. Stage 1 is a binary span extractor ("is this a substance-use impact?"); Stage 2 classifies the extracted span as Clinical or Social using a separate head. The two stages are trained independently.

Sentence + Token Hierarchy. A single model with two heads trained jointly: one for sentence presence, one for token BIO.

Span-nested GLiNER. A span scorer with explicit nested-span enumeration and overlap-aware decoding at inference.

Domain-adapted backbones. SocBERT (Reddit-domain) and StressRoBERTa (mental-health) are swapped in under the transformer token-classifier template while keeping the same optimiser, LR sweep, and training schedule.

5.2.4 Bucket 4: Ensemble techniques

Combining trained models: weight-space averaging (model soup) and prediction-space combination (majority vote / probability averaging).

Weight-space model soup. For a given family we fine-tune at every (lr, seed) configuration and average the weights of the top-K dev-F1 checkpoints: $\bar{\theta} = \frac{1}{K}\sum_k \theta_k$. This is only defined for same-architecture runs, so GLiNER, BiLSTM–CRF, and every pipeline family above are *not* soupable.

Prediction-space ensemble search. We run two complementary exhaustive searches, both over size-2–5 subsets:

- **Majority vote (cross-regime).** For each regime we take the top-7 relaxed-F1 runs and the top-7 strict-F1 runs, yielding up to 28 unique candidates. We then exhaustively evaluate all 122,409 size-2–5 subsets, picking the BIO tag for each token by hard majority. Because the voting operates on final BIO sequences, structured pipelines and GLiNER variants can participate.
- **Probability average (raw only).** Seven candidates, 112 size-2–5 subsets. Averaging is performed on per-token class distributions after re-aligning to the original token grid, so GLiNER span models and every structured pipeline—which do not expose comparable per-token tensors—are excluded from the candidate pool.

6 Experiments

Setup. All transformer experiments share the same optimiser defaults—AdamW with weight decay 0.01, 10% linear warmup then linear decay, dropout 0.1, max sequence length 512, and gradient clipping at 1.0. DeBERTa-family recipes are swept over lr $\in \{5e{-}4, 2e{-}5, 5e{-}5\}$ and GLiNER over $\{5e{-}6, 1e{-}5, 2e{-}5\}$ in both preprocessing regimes. Epochs, batch sizes, and recipe-specific knobs (e.g. SWA start epoch, FGM ε, gradient accumulation) vary by recipe and are listed in full in Appendix A.

Protocol. Innovations are compared *within* a regime first; the only stage that mixes regimes is the cross-regime ensemble search (§5.2.4). All tables below report dev-set F1; no test labels are used for selection.

6.1 Baseline model comparison

Table 1 compares the five model families under the raw-vs-preprocessed axis. BiLSTM–CRF and GLiNER gain from preprocessing; plain DeBERTa is stronger on raw text. Both domain-adapted encoders sit well below DeBERTa even in their own preferred regime: a Reddit- or mental-health-adapted pre-training corpus does not translate into a stronger BIO tagger for impact extraction.

Model	Raw	Prep
BiLSTM–CRF	0.251	0.296
GLiNER (fine-tuned)	0.463	0.524
SocBERT (Reddit)	–	0.404
StressRoBERTa (mental-health)	–	0.482
DeBERTa-v3-large baseline	**0.595**	0.545

Table 1: Model family comparison, relaxed F1 on dev. Dashes indicate configurations not run due to compute budget; SocBERT and StressRoBERTa were evaluated only in the preprocessed regime. GLiNER zero-shot scores 0.320 and is omitted for brevity.

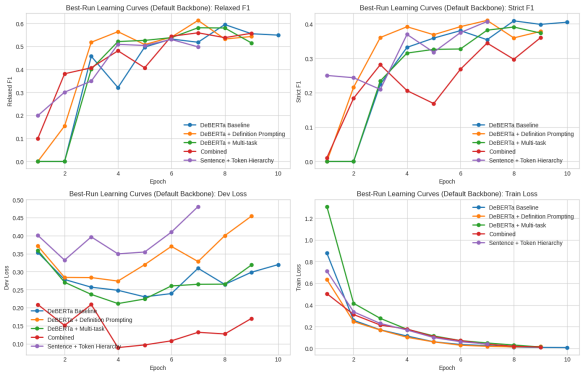

Figure 2: Best-run learning curves (preprocessed regime). **Top row:** Definition prompting (orange) tops relaxed F1 from epoch 4 onward, the clearest single-recipe win; strict F1 is within noise of the baseline. **Bottom row:** dev loss separates recipes even when F1 curves look close; the Sentence + Token Hierarchy drifts up after epoch 6, signalling mild overfit on this small training set.

6.2 Per-bucket ablations

Table 2 reports the per-bucket effects on top of the DeBERTa baseline, with relaxed/strict F1 shown in both regimes. The pattern is striking: only definition prompting (preprocessed regime, 0.613) clears the best baseline regime (0.595 raw). Several recipes improve on the preprocessed baseline (0.545) alone—R-Drop, multi-task, recall-boost, FGM + SWA—but none of them crosses the raw baseline, so in isolation they are strictly worse than simply running the plain encoder on raw text.

The learning-curve panel in Figure 2 makes the effect visible: definition prompting and the plain

Recipe	Raw (R / S) Relaxed	Raw (R / S) Strict	Preprocessed (R / S) Relaxed	Preprocessed (R / S) Strict
DeBERTa baseline	0.595	0.409	0.545	0.421
+ Focal loss	0.541	0.341	0.538	0.358
+ Multi-task	0.581	0.382	0.550	0.394
+ **Definition prompting**	0.575	0.350	**0.613**	**0.410**
+ Recall-Boost (seed 42)	0.582	0.312	0.560	0.330
+ R-Drop (seed 123)	0.561	0.360	0.577	0.380
+ FGM + SWA (seed 42)	0.573	0.356	0.547	0.256

Table 2: Per-bucket effect on top of the DeBERTa baseline. All numbers are dev F1; best LR per family is selected from $\{5e{-}4,\ 2e{-}5,\ 5e{-}5\}$ within each regime. Only definition prompting (preprocessed) clears the best baseline regime of 0.595 relaxed F1.

Pipeline	Raw (R / S)	Prep (R / S)
Hierarchical DeBERTa	**0.604** / 0.372	0.559 / 0.383
Two-Step Impact	–	0.534 / 0.404
Sentence + Token	–	0.530 / 0.375
Span-nested GLiNER	–	0.443 / 0.491

Table 3: Structured prediction pipelines (dev F1). Only Hierarchical DeBERTa was run under both regimes; dashes indicate configurations not run due to compute budget.

Ensemble	Relaxed	Strict	Prec.	Rec.
2 models (majority)	0.623	0.465	0.586	0.664
3 models (majority)	0.621	0.507	0.638	0.605
4 models (majority)	**0.635**	0.509	0.648	0.623
5 models (majority)	0.631	**0.518**	0.670	0.597
Prob. avg. (raw, 7 cand.)	0.614	0.431	0.653	0.580

Table 4: Ensemble search. Cross-regime majority vote across $122{,}409$ size-2–5 subsets; probability average across 112 size-2–5 subsets in the raw regime only (span/pipeline families are excluded because they do not emit per-token distributions).

baseline plateau at ≈ 0.60 relaxed F1, while other recipes oscillate around or below it. Dev loss is a noisier signal than F1 and sometimes separates recipes that F1 treats as similar (the Sentence + Token Hierarchy variant drifts up after epoch 6, which the raw F1 number does not show).

6.3 Structured prediction pipelines

Table 3 compares the four structured variants. Hierarchical DeBERTa is the clear winner and becomes the anchor of our final ensemble. Strict F1 on the hierarchical pipeline actually goes up under preprocessing even though relaxed F1 goes down: the sentence-gating head tightens predicted boundaries when the token head is working with cleaner input. The Two-Step Impact Pipeline trades recall for precision and reaches the highest strict F1 of any single-model recipe (0.404).

6.4 Model soup

The best soup averages the top-3 checkpoints of the Recall-Boost seed-42 sweep and reaches relaxed F1 $= 0.595$ (strict 0.331) — matching the plain baseline but not exceeding it, consistent with Wortsman et al. (2022): soup helps competitive recipes and does not rescue weaker ones.

6.5 Ensemble search

Table 4 shows the top ensemble sizes from the cross-regime majority-vote search, plus the best probability-average result. The four-model majority ensemble reaches relaxed F1 $=$ **0.635**, a $+0.022$ jump over the best single model and $+0.025$ over the published baseline on the same dev split. The strict-F1 optimum is a different five-model subset at 0.518. Hard voting consistently beats soft averaging on this benchmark (0.635 vs 0.614), which is a reversal of the usual deep-learning preference for logit averaging.

The winning four-model subset pairs the Hierarchical DeBERTa pipeline as an anchor with three complementary DeBERTa recipes drawn from both preprocessing regimes: a definition-prompting run from the preprocessed regime and two raw-regime runs with different regularisation objectives. The ensemble benefits from both *preprocessing diversity* (members trained on different input distributions) and *objective diversity* (definition prompting, multi-task, sentence gating), which is where the $+0.022$ headroom over the best single model comes from. The full LR-selected ablation in one view (BiLSTM–CRF, GLiNER variants, domain backbones, all DeBERTa recipes, pipelines, soup, ensemble, and published base-

Rank	Team / System	Strict	Relaxed
1	*top team*[*]	0.53	0.60
2	*runner-up*[*]	0.51	0.60
3	*third place*[*]	0.5	0.58
5	**Ours (submitted run)**	**0.49**	**0.45**
–	Median across 11 teams[*]	0.48	0.58

Table 5: SMM4H–HeaRD 2026 Task 7 test-set leaderboard (Lopez-Garcia et al., 2026). Rows marked [*] are placeholders pending release of the official overview paper. Our submitted system is an earlier configuration than the dev-best four-model ensemble of Table 4; that ensemble was finalised after the deadline and was not evaluated on test.

lines) is deferred to Appendix B (Table 6).

6.6 Shared-task leaderboard

We submitted to the SMM4H–HeaRD 2026 Task 7 official leaderboard (Lopez-Garcia et al., 2026) and placed **5th of 11** participating teams, with test-set strict F1 $= 0.49$ and relaxed F1 $= 0.45$. Table 5 summarises the leaderboard in context. Two caveats apply to our submitted number. First, the system we submitted was an *earlier* configuration than the four-model majority-vote ensemble reported in Table 6; the ensemble recipe was finalised after the submission window closed, so its test-set score is unmeasured. Second, on the dev split the relaxed>strict ordering holds as expected (0.635 vs 0.518); on test the ordering flips (0.45 vs 0.49), which is consistent with the train→test label prior flip discussed in §9: the test split is dominated by longer, multi-token Social spans, so strict boundary matches are more frequent than the dev-set ordering would predict.

7 Error Analysis

We ran a targeted error analysis on the best relaxed single model (definition prompting, preprocessed regime, lr $= 5\text{e}{-5}$, relaxed F1 $= 0.613$, strict F1 $= 0.410$). Four categories dominate:

Boundary errors. The ≈ 0.20 gap between relaxed and strict F1 is almost entirely attributable to partial span matches on multi-word Social entities. Phrases such as "charged with disorderly conduct" are typed correctly but the predicted span misses the "charged with" prefix.

Implicit false negatives. Single-token implicit Clinical expressions ("hell", "broken", "numb") and long-tail colloquial phrases are the dominant category of missed spans. These spans do not contain any clinical vocabulary and the model falls back to the "no-entity" prior.

Negation handling. Negated phrases are mostly rejected correctly; type confusion between Clinical and Social is rare.

Third-person references. The model over-rejects spans that describe someone else's impacts rather than the poster's, despite no explicit training signal for this distinction.

These categories suggest three concrete future-work directions: boundary-aware decoding for partial matches, targeted synthetic augmentation for implicit single-token expressions, and a perspective-tagging auxiliary signal to separate first-person from third-person mentions. Negation handling is already reliable and does not need dedicated modelling on this benchmark.

8 Discussion and Key Insights

Our headline finding is that *training-objective diversity* is a much stronger lever on this benchmark than individual-model polish. A four-model majority-vote ensemble of relatively ordinary recipes beats any single one of them by $+0.022$ relaxed F1. Three secondary observations are worth calling out:

Learning rate must be swept per family. The best LR is family-specific: $2\text{e}{-5}$ for the plain baseline, $5\text{e}{-5}$ for definition prompting and multi-task. Running a single global LR would have reversed the ordering between at least three recipes and hidden definition prompting's improvement.

Domain pre-training does not help here. SocBERT and StressRoBERTa, both plausible a priori, underperform general-purpose DeBERTa by ≈ 0.09–0.19 relaxed F1. For a task whose labels are semantic categories defined at annotation time, generic language understanding appears to matter more than register-specific pre-training.

Majority vote beats probability average. On our dev set hard voting is $+0.021$ better than soft averaging (0.635 vs 0.614). We hypothesise that the most confident recipes are also the most over-confident: their logit mass dominates the average and propagates their particular mistakes, whereas a hard vote caps each member's influence at one. On a small, noisy benchmark with decorrelated er-

rors, this cap is worth more than the information lost by discretising the output.

9 Limitations

Dev-only evaluation. We report dev-set performance throughout. Test labels were not available during the study, and the dev split itself is only 258 documents, so a confidence interval on relaxed F1 is wide. Given the $75\% \rightarrow 70\%$ label prior flip, test numbers may well shift from what we report; the system rankings we report are meaningful *within* the dev distribution but do not guarantee test-set generalisation.

No cross-post context. Our models see one post at a time. User history within a subreddit is a potentially strong signal, particularly for disambiguating third-person references, and we do not exploit it.

Ensemble cost. The best system runs four models at inference time and pays a $\approx 4\times$ latency penalty relative to a single-model deployment. In practice the hierarchical pipeline at 0.604 captures ≈ 0.03 below the ensemble at a single-model cost.

Coverage of domain encoders. Budget constraints restricted SocBERT and StressRoBERTa to the preprocessed regime only. It is possible—but in our view unlikely given the raw/prep ordering of the other transformers—that a raw run of either encoder would close the gap to DeBERTa.

Submitted system $\neq$ dev-best system. The leaderboard number in §6.6 was produced by an earlier configuration; the four-model ensemble with the best dev performance was assembled after the competition deadline. Our test-set evidence is therefore for a weaker member of the design space we explore, and the ensemble's actual test performance remains unverified.

10 Conclusion

We presented a systematic study of the Reddit-Impacts impact-extraction task, organised as four contribution buckets (imbalanced-inputs training, advanced regularisation, architecture, and ensemble techniques)—most layered on a DeBERTa backbone—evaluated under a consistent LR sweep and a paired raw-vs-preprocessed protocol that runs orthogonally to the buckets. Our best single model, definition prompting, reaches relaxed F1 $= 0.613$. Our best system, a four-model majority-vote ensemble, reaches 0.635 and exceeds the published baseline by $+0.025$ on the same dev split. On the official SMM4H–HeaRD 2026 Task 7 leaderboard (Lopez-Garcia et al., 2026) our submitted system placed 5th of 11 teams (strict F1 $= 0.49$, relaxed F1 $= 0.45$) using an earlier configuration; the dev-best ensemble was finalised after the deadline and is unverified on test. The pattern across buckets is consistent: no single regularisation trick beats a well-tuned DeBERTa, and the headroom comes from combining decorrelated recipes via hard voting. We see the ≈ 0.18 gap to the human ceiling as coming mainly from boundary decoding and implicit-expression recall, and view these as the most promising future-work directions.

Acknowledgments

We thank the CE7455 course staff at NTU, and the SMM4H–HeaRD 2026 shared-task organisers for releasing Reddit-Impacts 2.0.

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A Hyperparameters and Training Setup

All transformer runs use AdamW with $\beta = (0.9, 0.999)$, weight decay 0.01, max sequence length 512, dropout 0.1, linear warmup over 10% of total steps, and linear decay, with gradient clipping at 1.0. The DeBERTa LR sweep is $\{5\mathrm{e}{-4}, 2\mathrm{e}{-5}, 5\mathrm{e}{-5}\}$; the GLiNER sweep is $\{5\mathrm{e}{-6}, 1\mathrm{e}{-5}, 2\mathrm{e}{-5}\}$. Per-recipe schedules:

- DeBERTa baseline, focal, definition prompting, multi-task, Recall-Boost, and domain backbones (SocBERT, StressRoBERTa): 30 epochs at batch size 30.
- R-Drop: 90 epochs at physical batch 8 with gradient accumulation $\times 4$ (effective batch 32), LR $1.5\mathrm{e}{-5}$.

Model	Relaxed	Strict	LR
BiLSTM–CRF	0.296	0.256	5e-4
GLiNER (fine-tuned)	0.524	–	2e-5
Span-nested GLiNER	0.443	–	1e-5
SocBERT	0.404	0.277	5e-5
StressRoBERTa	0.482	0.270	2e-5
DeBERTa baseline	0.595	0.409	2e-5
+ Focal loss	0.541	0.341	2e-5
+ Multi-task	0.581	0.382	5e-5
+ Definition prompting	**0.613**	0.410	5e-5
Hierarchical DeBERTa	0.604	0.372	2e-5
Two-Step Impact	0.534	0.404	2e-5
Sentence + Token	0.530	0.375	5e-5
Recall-Boost (s42)	0.582	0.312	2e-5
R-Drop (s123)	0.577	0.380	2e-5
FGM + SWA (s42)	0.573	0.356	2e-5
Model soup (recall-boost s42)	0.595	0.331	2e-5
4-model majority ensemble	**0.635**	0.509	–
GPT-4o 3-shot (reported)	*0.440*	–	–
DeBERTa published SOTA	*0.610*	–	–

Table 6: Full LR-selected ablation (dev F1). Strict F1 for GLiNER variants is not exposed by the toolkit. GPT-4o 3-shot and DeBERTa SOTA numbers are reported from Dey et al. (2025) on the same dev split.

- FGM + SWA: 120 epochs at physical batch 8 with gradient accumulation $\times 4$, LR $1.5\mathrm{e}{-5}$, $\varepsilon = 0.5$, SWA averaging from epoch 10.
- Hierarchical DeBERTa, Two-Step Impact, Sentence + Token: 10 epochs at batch size 16.
- BiLSTM–CRF: 30 epochs at batch size 30, AdamW with base LR $1\mathrm{e}{-3}$, frozen GloVe-300d, 2-layer bidirectional LSTM (2×256).
- GLiNER fine-tune / span-nested: 5 epochs at batch size 8, 0.4 span-score threshold, following the default schedule of Zaratiana et al. (2024).

B Full LR-Selected Ablation

Table 6 presents the full LR-selected ablation in one view. BiLSTM–CRF sits at 0.296; domain-adapted encoders underperform the general-purpose DeBERTa; the only recipe that improves on the baseline alone is definition prompting; the four-model ensemble sits clearly above the published SOTA on the same split.

C Reproducibility

All experiments use PyTorch with mixed-precision training on a single NVIDIA A100 GPU. The cross-regime ensemble search runs as a pure CPU post-processing step on cached BIO sequences from each candidate and takes roughly 45 minutes for all 122,409 size-2–5 subsets. The probability-average search takes $\approx$ 5 minutes on the 112 candidate subsets.